

Canakit Raspberry Pi 3

LESSON · GRADES 6-8

Minecraft Python Mods

LEARNING OUTCOMES

- Students will modify and run Python scripts that alter a Minecraft world
- Students will use coordinates to position objects in 3D space
- Students will explain the relationship between code and its visible effects

MATERIALS

- Raspberry Pi kits connected to monitors, keyboards, mice
- Minecraft Pi + Python (preinstalled on Raspberry Pi OS)
- Printed starter scripts / cheat sheet of mcpi commands

INSTRUCTIONS

1. Set up stations in advance and verify Minecraft Pi launches.
2. Demo the magic moment: a one-line Python command teleports the player.
3. Students run the starter script, then modify coordinates and block types.
4. Challenge ladder: place a block → build a wall with a loop → build a house function.
5. Free-build challenge: auto-construct something using at least one loop.
6. Screenshots + code snippets go in a shared doc as the deliverable.

ACTIVITY

Minecraft Python Mods — students manipulate a world they care about with real text-based code. Loops stop being abstract when one loop builds a castle wall.

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LESSON · GRADES 9-12

Build a Web Server

LEARNING OUTCOMES

- Students will configure a Raspberry Pi as a functioning web server
- Students will explain the request-response cycle between client and server
- Students will author and serve a custom HTML page over a local network

MATERIALS

- Raspberry Pi kits with monitors and peripherals
- Second device per group to act as the 'client'
- Printed setup steps for installing/enabling a web server (e.g., `python -m http.server` or Apache)
- HTML starter template

INSTRUCTIONS

1. Groups boot the Pi and find its IP address on the local network.
2. Start a simple web server and confirm the default page loads from the Pi itself.
3. Write a custom HTML page (bio, topic page, or joke site) and serve it.
4. The payoff: load the page from a different device using the Pi's IP.
5. Diagram the round trip: browser request → server → response → render.
6. Discussion: what's the same and different about how google.com reaches you?

ACTIVITY

Build a Web Server — the moment another device loads 'their' page cold is when the internet stops being magic. Groups finish by diagramming exactly what traveled over the wire and why.